

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)

Assessment Tool Weightage: 50%

QUESTIONS: 5

Duration : 1 Hour

Total Marks:50

Annexure A

Analytical Problem Solving

Code of Conduct:

I **M.VAISHNAV (192524251)** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

M.VAISHNAV (192524251)

Signature of the student:

1. Water Jug Problem (4-Gallon Jug & 3-Gallon Jug)

Initial State

(4-gallon jug = 0, 3-gallon jug = 0)

Goal

Obtain exactly **2 gallons** in the **4-gallon jug**.

Step 4-Gallon Jug 3-Gallon Jug			Action
1	0	3	Fill 3-gallon jug
2	3	0	Pour into 4-gallon jug
3	3	3	Fill 3-gallon jug again
4	4	2	Pour into 4-gallon jug until full
5	0	2	Empty 4-gallon jug
6	2	0	Pour remaining 2 gallons into 4-gallon jug

Goal Achieved

4-gallon jug contains exactly 2 gallons.

2. Mars Rover Agent Analysis

(i) Percepts

The rover receives:

- Camera images
- Soil composition readings
- Rock sample data
- Temperature measurements
- Atmospheric conditions
- Terrain and obstacle information
- GPS/location data

(ii) Operating Environment

Characteristic	Type
Observability	Partially Observable
Determinism	Stochastic
Episodic/Sequential	Sequential
Static/Dynamic	Dynamic
Discrete/Continuous	Continuous
Single/Multi Agent	Single Agent

(iii) Actions

- Move forward/backward
- Turn left/right
- Avoid obstacles
- Collect rock samples
- Drill soil
- Capture images
- Transmit data to Earth
- Analyze samples

(iv) Performance Evaluation

The rover is evaluated based on:

- Number of successful samples collected
- Scientific discoveries
- Accuracy of analysis
- Energy efficiency
- Distance covered
- Safe navigation
- Mission completion rate

(v) Suitable Agent Architecture

Model-Based Goal-Oriented Agent

Reason:

- Maintains internal map of Mars.
 - Plans routes toward scientific targets.
 - Adapts to uncertain environments.
 - Uses sensor information continuously.
 - Makes autonomous decisions.
-

3. 8-Queens Problem Formulation

Problem Statement

Place 8 queens on a chessboard so that no two queens attack each other.

State Space

A state is a chessboard configuration with queens placed.

Example:

Q _ _ _ _ _ _ _

_ _ Q _ _ _ _ _

_ _ _ _ Q _ _ _

...

Initial State

Empty board with no queens.

Actions

Place a queen in any safe square of the next column.

Transition Model

Adding one queen to a safe position creates a new state.

Goal Test

A solution is reached when:

- 8 queens are placed
 - No two queens share:
 - Same row
 - Same column
 - Same diagonal
-

Path Cost

Each queen placement costs 1.

Total cost = 8.

Search Strategy

Use:

- Depth First Search (DFS)
 - Backtracking Search
-

Example Solution

Column Row

1	1
2	5
3	8
4	6
5	3
6	7
7	2
8	4

This arrangement satisfies all constraints.

4. OLA Cab Booking Application

Suitable Problem-Solving Agent

Goal-Based Agent

Reason:

The customer's goal is:

"Reach destination using the most suitable cab."

The agent evaluates options and chooses a cab that satisfies the goal.

Pseudocode

```
BEGIN

INPUT source
INPUT destination

DISPLAY available cab types
(Mini, Micro, Sedan, Shared, Prime)

USER selects cab

IF cab available THEN
    Calculate fare
    Display fare and ETA
    Confirm booking
    Assign driver
    Start trip
ELSE
    Display "Cab Not Available"
ENDIF

END
```

Working

1. User enters source and destination.
2. System searches available cabs.
3. User selects preferred cab.
4. Fare is calculated.
5. Driver is assigned.
6. Trip begins.

5. Uniform Cost Search (UCS) – Logistics Network

The graph image is not visible in the uploaded PDF, so the warehouse connections and route costs are missing.

UCS Algorithm

```
UniformCostSearch(start, goal)
```

1. Create Priority Queue
2. Insert start node with cost 0
3. While queue is not empty
 - Remove node with lowest cost
 - If node = goal
Return solution
 - Expand node
 - For each child
 - New Cost =
Current Cost + Edge Cost
 - Insert child into queue
4. End While
5. Return Failure

UCS Properties

- Expands lowest-cost node first.
- Guarantees optimal solution.
- Uses Priority Queue.
- Complete and Optimal when costs are positive.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets the problem and identifies all parameters and requirements accurately.	Understands the problem with minor gaps.	Partial understanding; misses key elements.	Misinterprets or is unable to understand the problem.
Method Selection	20%	Selects the most appropriate method/formula with strong justification.	Selects the correct method with minor errors.	Uses an inappropriate or partially correct method.	Unable to select an appropriate method.
Solution Process	25%	Logical and systematic steps with the correct approach throughout.	Mostly correct steps with minor mistakes.	Disorganized steps with multiple errors.	No clear solution process.
Accuracy of Calculation	20%	All calculations are accurate and the final answer is correct.	Minor calculation errors.	Frequent errors leading to an incorrect final answer.	No valid calculations.
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance.	Basic interpretation with minor omissions.	Limited or unclear interpretation.	No interpretation provided.